United Arab Emirates University

College of Engineering &
College of Science

Math 2220: Linear Algebra and Engineering Applications

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ENGINEERING APPLICATIONS OF SYSTEMS OF LINEAR EQUATIONS

Many "real life" situations are governed by a system of differential equations. Physical problems usually have more than one dependent variable to be considered. Models of such problems can give rise to system of differential equations in which there are two or more dependent variable and one independent variable.

Because we are going to be working almost exclusively with systems of equations in which the number of unknowns equals the number of equations we will restrict our discussion to these kinds of systems.

Let's start with the following system of *n* equations with the *n* unknowns, $x_1, x_2, \ldots, x_n$

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n
\end{aligned}
$$

Note that in the subscripts on the coefficients in this system, a_{ij}, the *i* corresponds to the equation that the coefficient is in and the *j* corresponds to the unknown that is multiplied by the coefficient. This can be put in a simple linear equation form

$$ax = b,$$

Where *a* and *b* are given real or complex numbers and *x* is an unknown.

To use linear algebra to solve this system we will first write down the **augmented matrix** for this system. An augmented matrix is really just the all the coefficients of the system and the numbers for the right side of the system written in matrix form. Here is the augmented matrix for this system. However in our analysis we will use Matlab to help us solve such system.

$$
\begin{pmatrix}
a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\
a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\
\vdots & \vdots & \ddots & \vdots & \vdots \\
a_{n1} & a_{n2} & \cdots & a_{nn} & b_n
\end{pmatrix}
$$

We will start with the simplest possible linear equation $ax = b$, such equation has three possible solutions.

a) If $a = 0$ and $b \neq 0$, then $0x = 0 = b$ which is false for any value of *x*, and so there are no solutions
b) If $a \neq 0$, then the equation has a unique solution $x = b/a$ for any value of *b*.

c) If $a = b = 0$, then $0x = 0$ for any value of x and so there are infinitely many solutions.

General Strategy for Solving Systems of Linear Equations:
Using Mathematic to solve an applied problem involves translation of the features of the problem into mathematical language (terminology, symbols, equations and so on). We refer to this translation process as building a mathematical model of the problem.

In all applications of linear equations, we will follow the same general strategy

First: ***Identify and label the unknowns.***
In other words, what are we asked to find? In answering this question, you should note down something like the following:

Let x be the number of video games.
Let y be the number of applications.
Let z be the number of documents.

Note that all the unknowns should be *numbers*, so we should *not* say something like "Let x = video games."

Second: ***Use the information given to set up equations in the unknowns.***
How to do this depends on the way the problem is worded. We will look at a few examples below to develop some strategies.

Third: ***Solve the system to obtain the values for the unknowns.***
We will use the help of Matlab to obtain solve for the unknowns.

There are several kinds of applications generally found in textbooks.

- Applications in which the given information can be tabulated
- Applications in which some of the given information must be translated from words into equations.
- Applications of specialized types, such as "transportation problems" and "traffic flow problems." These require special techniques for setting up the system of equations.

We will start with simple applications and then we will move on into more specialized Engineering applications.

Example 1:
A Yogurt Company makes three yogurt blends: LimeOrange, using 2 quarts of lime yogurt and 2 quarts of orange yogurt per gallon; LimeLemon, using 3 quarts of lime yogurt and 1 quart of lemon yogurt per gallon; and OrangeLemon, using 3 quarts of orange yogurt and 1 quart of lemon yogurt per gallon. Each day the company has 800 quarts of lime yogurt, 650 quarts of orange yogurt, and 350 quarts of lemon yogurt available. How many gallons of each blend should it make each day if it wants to use up all of the supplies?

The unknowns in the problem are related to what?

The number of gallons of each yogurt blends

How many unknowns in the problem?

The three unknowns can be specified as follows:

Let x = the number of gallons of LimeOrange the company should make
Let y = the number of gallons of LimeLemon the company should make
Let z = the number of gallons of OrangeLemon the company should make

You can also list these in some other order, for instance:

Let x = the number of gallons of LimeLemon the company should make
Let y = the number of gallons of LimeOrange the company should make
Let z = the number of gallons of OrangeLemon the company should make.

Note For the rest of the problem, we shall assume the first ordering.

We can organize the given information in a table. To set up the table, do the following:

- Place the categories corresponding to the unknowns along the top.
- Add an extra column for the "Total Available"

	LimeOrange	LimeLemon	OrangeLemon	Total Available
Lime Yogurt (quarts)				
Orange Yogurt (quarts)				
Lemon Yogurt (quarts)				

Now read across the first row of the table: it gives the amounts of lime yogurt needed for the three blends, and also the total available.

If the company makes x quarts of LimeOrange, y quarts of LimeLemon, and z quarts of OrangeLemon, it will need a total of

$$2x + 3y$$

quarts of lime yoghurt. Since Softflow has a total of 800 quarts of lime yogurt on hand, and it wants nothing left over, we must have

Amount used = Amount Available

$$2x + 3y = 800$$

Similarly, we get two more equations for orange and lemon yogurt:

Q Equation for Orange Yogurt: $2x + 3z = 650$

Q Equation for Lemon Yogurt: $y + z = 350$

Now you have a system of three equations in three unknowns.

$$\begin{bmatrix} 2 & 3 & 0 \\ 2 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 800 \\ 650 \\ 350 \end{bmatrix}$$

This system of equations can be easily solved by Matlab.

Example 2:

Last year you purchased shares in three Internet companies: OHaganBooks.com, FarmersBooks.com, and JungleBooks.com. The OHaganBooks.com cost you $50 per share, FarmersBooks.com stocks cost you $45 per share, and JungleBooks.com cost you $30 per share. You spent a total of $24,400, and purchased twice as many FarmersBooks.com shares as JungleBooks.com. The OHaganBooks.com stocks appreciated by 20%, while the other two appreciated by 10%, and you sold all the stocks for $3,440 more than you originally paid. How many stocks of each company did you originally purchase?

The unknowns in the problem are related to what?

How many unknowns in the problem?

The three unknowns can be specified as follows:

Let x = the number of stocks of OHaganBooks.com you purchased.
Let y = the number of stocks of FarmersBooks.com you purchased.
Let z = the number of stocks of JungleBooks.com you purchased.

You can also list these in some other order, for instance:

Let x = the number of stocks of FarmersBooks.com you purchased.
Let y = the number of stocks of JungleBooks.com you purchased.
Let z = the number of stocks of OHaganBooks.com you purchased.

Note For the rest of the problem, we shall assume the first ordering.

Now look at the first piece of information you are given:

The OHaganBooks.com cost you $50 per share, FarmersBooks.com stocks cost you $45 per share, and JungleBooks.com cost you $30 per share. You spent a total of $24,400.

$50x + 45y + 30z = 24{,}400$

Now look at the next piece of information:

[You] purchased twice as many FarmersBooks.com shares as JungleBooks.com.

The best way to translate this into an equation is to *reword the given information in such a way as it contains the phrases for the relevant unknowns: "the number of shares of FarmersBooks.com", and "the number of shares of JungleBooks.com".*

If you rewrite the sentence using these phrases, you get:

The number of shares of FarmersBooks.com was twice the number of shares of JungleBooks.com

In symbols, this is:

$y = 2z.$

Now look at the third piece of information:

The OHaganBooks.com stocks appreciated by 20%, while the other two appreciated by 10%, and you sold all the stocks for $3,440 more than you originally paid.

Q Select which (if any) of the following equations conveys this information.

A $0.20x + 0.10y + 0.10z = 3{,}440$ **B** $20x + 10y + 10z = 3{,}440$
C $10x + 4.5y + 3z = 3{,}440$ **D** $5.5x + 2.5y + 3.5z = 3{,}440$

Let us look at the OHaganBooks.com stock first: You purchased x shares at $50 each, costing you a total of $50x. Since the stocks appreciated by 20%, the profit earned is 20% of the purchase price, or

$0.2 \times 50x = 10x.$

Similarly, the FarmersBooks.com shares cost you 45y, and appreciated by 10%, or

$0.1 \times 45y = 4.5y.$

Finally, the JungleBooks.com shares cost you 30z, and appreciated by 10%, or

$0.1 \times 30z = 3z.$

Adding all the progits together gives:

$10x + 4.5y + 3z = 3{,}440$

Now you have a system of three equations in three unknowns:

$$\begin{aligned} 50x + 45y + 30z &= 24{,}400 \\ y - 2z &= 0 \\ 10x + 4.5y + 3z &= 3{,}440 \end{aligned}$$

In Class Exercise 1:

A logging company has a contract with a local mill to provide **1000 m³** of *Lodgepole pine*, **800 m³** of *spruce*, and **600 m³** of *Douglas fir* logs per month. There are three regions available for logging. The following table gives the species mix, and timber density for each region.

Region	Volume/hectare	% Pine	% Spruce	% Fir
West	330 m^3 / ha	70 %	20 %	10 %
North	390 m^3 / ha	10 %	60 %	30 %
East	290 m^3 / ha	5 %	20 %	75 %

How many hectares should one log in each operating region listed above to deliver exactly the required volume of logs? I don't want to have to store logs so I don't want any left over at the end of each month, but I do need to make my quota.

Answer:

- x = number of hectares logged in West region
- y = number of hectares logged in North region
- z= number of hectares logged in East region

x = 3.92, y = 2.05, and z = 1.06

Things to note:

Whenever you model a system of equation see if number of equations and number of unknowns follow one of the following conditions. If

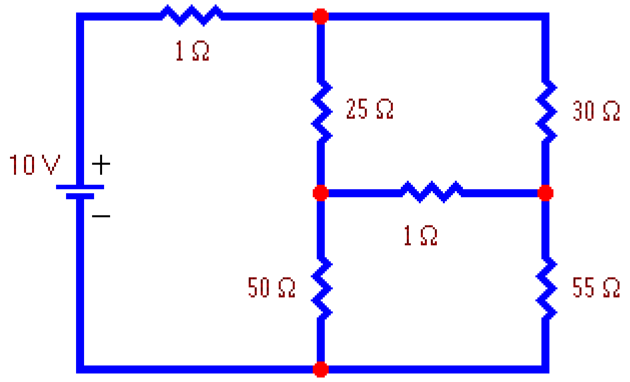
a) Number of equations is more than the number of unknowns, and then the system is over-determined and, in general, will not yield a solution.

b) Number of equations is less than the number of unknowns, and then we have *too few* equations to determine a unique solution. The system is said to be indeterminate. To actually solve for the system, other properties of the system should be considered, such as the material properties.

c) Number of equations equal to the number of unknowns, the system is will defined and a solution can be obtained.

LINEAR ALGEBRA IN ELECTRICAL ENGINEERING APPLICATION

One of the most important applications of linear algebra to electronics is to analyze electronic circuits that cannot be described using the rules for resistors in series or parallel such as the one shown to the right. The goal is to calculate the current flowing in each branch of the circuit or to calculate the voltage at each node of the circuit.



Knowing the branch currents, the nodal voltages can easily be calculated, and knowing the nodal voltages, the branch currents can easily be calculated. Loop analysis finds the currents directly and nodal analysis finds the voltages directly. Which method is simpler depends on the given circuit. Nodal analysis is important because its answers can be directly compared with voltage measurements taken in a circuit, whereas currents are not so easily measured in a circuit (one must cut wires). Click on one of the following topics:

Loop Current Analysis of Electric Circuits
Nodal Voltage Analysis of Electric Circuits

1. Loop Current Analysis of Electric Circuits

The steps in the loop current method are:

1. Count the number of required loop currents required. Call this number *m*.
2. Choose *m* independent loop currents, call them $I_1, I_2, \ldots, I_m$ and draw them on the circuit diagram.
3. Write down Kirchoff's Voltage Law for each loop. The result, after simplification, is a system of *n* linear equations in the *n* unknown loop currents in this form:

$$\begin{cases} R_{11}\cdot I_1 + R_{12}\cdot I_2 + \cdots + R_{1m}\cdot I_m = V_1 \\ R_{21}\cdot I_1 + R_{22}\cdot I_2 + \cdots + R_{2m}\cdot I_m = V_2 \\ \quad\vdots \qquad\quad \vdots \qquad\qquad\quad \vdots \qquad \vdots \\ R_{m1}\cdot I_1 + R_{m2}\cdot I_2 + \cdots + R_{mm}\cdot I_m = V_m \end{cases}$$

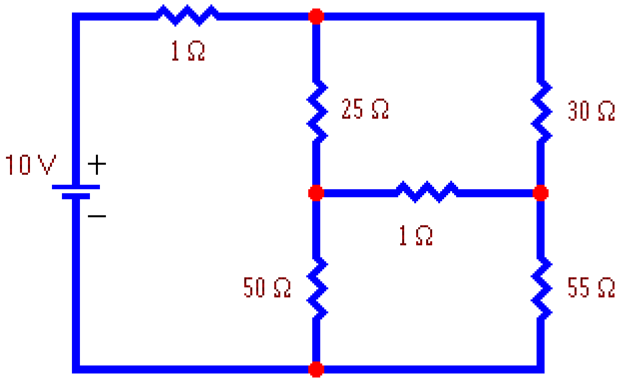
where $R_{11}, R_{12}, \ldots, R_{mm}$ and $V_1, V_2, \ldots, V_m$ are constants.

4. Solve the system of equations for the m loop currents $I_1, I_2, \ldots, I_m$ using Gaussian elimination or some other method.

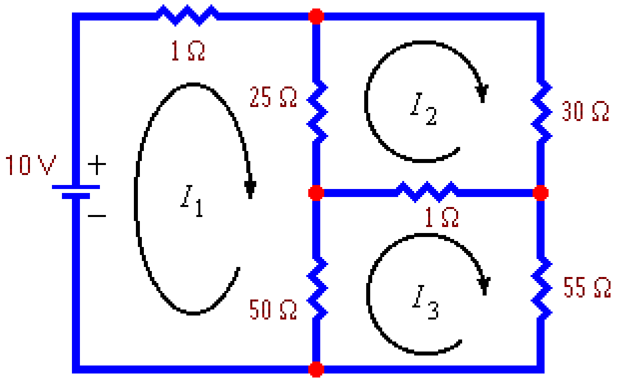
5. Reconstruct the branch currents from the loop currents.

Example 1:

Find the current flowing in each branch of this circuit.



Solution:
The number of loop currents required is 3.



We will choose the loop currents shown to the right. In fact these loop currents are **mesh currents**.

Write down Kirchoff's Voltage Law for each loop. The result is the following system of equations:

$$\begin{cases} 1i_1 + 25(i_1 - i_2) + 50(i_1 - i_3) = 10 \\ 25(i_2 - i_1) + 30i_2 + 1(i_2 - i_3) = 0 \\ 50(i_3 - i_1) + 1(i_3 - i_2) + 55i_3 = 0 \end{cases}$$

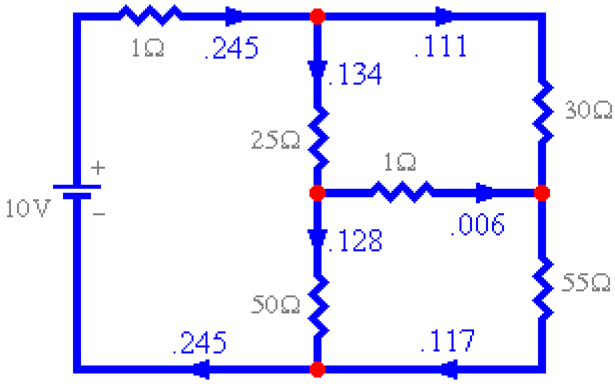
Collecting terms this becomes:

This form for the system of equations could have been gotten immediately by using the inspection method.

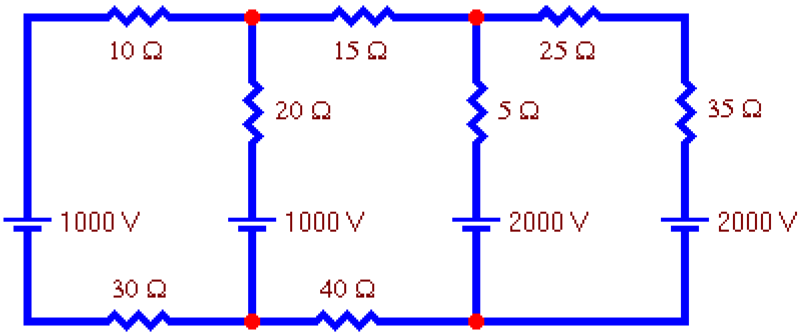
Solving the system of equations using Gaussian elimination or some other method

$$\begin{cases} 76i_1 - 25i_2 - 50i_3 = 10 \\ -25i_1 + 56i_2 - 1\, i_3 = 0 \\ -50i_1 - 1\, i_2 + 106i_3 = 0 \end{cases}$$

gives the following currents, all measured in amperes: I_1=0.245, I_2=0.111 and I_3=0.117

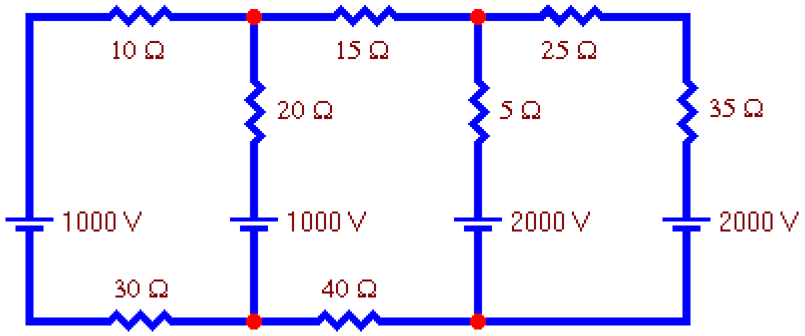


Reconstructing the branch currents from the loop currents gives the results shown in the picture to the right.



Exercise 1:

Find the current flowing in each branch of this circuit.


10 Ω
15 Ω
25 Ω
20 Ω
5 Ω
35 Ω
1000 V
1000 V
2000 V
2000 V
30 Ω
40 Ω

2. Nodal Voltage Analysis of Electric Circuits

In this method, we set up and solve a system of equations in which the unknowns are the **voltages at the principal nodes of the circuit**. From these nodal voltages the currents in the various branches of the circuit are easily determined. The steps in the nodal analysis method are:

Count the number of principal nodes or junctions in the circuit. Call this number n. (A **principal node or junction** is a point where 3 or more branches join. We will indicate them in a circuit diagram with a red dot. Note that if a branch contains no voltage sources or loads then that entire branch can be considered to be one node.)

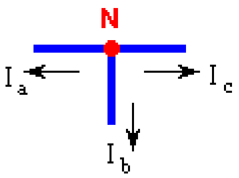
Number the nodes $N_1, N_2, \ldots, N_n$ and draw them on the circuit diagram. Call the voltages at these nodes $V_1, V_2, \ldots, V_n$, respectively.

Choose one of the nodes to be the reference node or ground and assign it a voltage of zero.

For each node except the reference node write down Kirchoff's Current Law in the form *"the **algebraic sum** of the currents flowing out of a node equals zero"*. (By algebraic sum we mean that a current flowing into a node is to be considered a negative current flowing out of the node.)

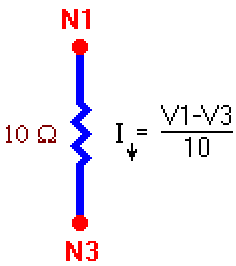
For example, for the node to the right KCL yields the equation:

$I_a + I_b + I_c = 0$

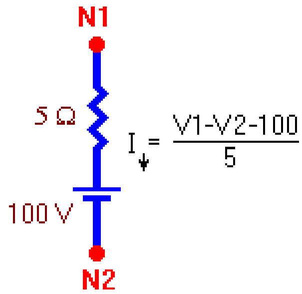


Express the current in each branch in terms of the nodal voltages at each end of the branch using Ohm's Law ($I = V / R$). Here are some examples:

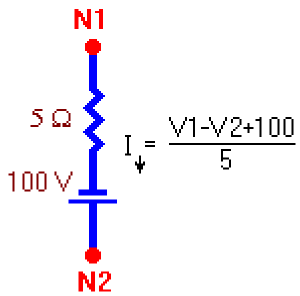
- The current downward out of node 1 depends on the voltage difference V1 - V3 and the resistance in the branch.



- In this case the voltage difference across the resistance is V1 - V2 **less the voltage across the voltage source**. Thus the downward current is as shown.



- In this case the voltage difference across the resistance must be 100 volts greater than the difference V1 - V2. Thus the downward current is as shown.



The result, after simplification, is a system of m linear equations in the m unknown nodal voltages (where m is one less than the number of nodes; $m = n - 1$). The equations are of this form:

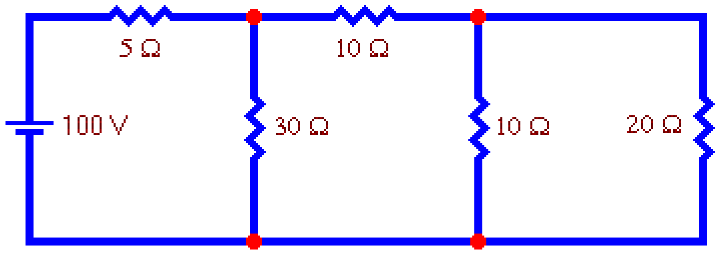
$$\begin{cases} G_{11} \cdot V_1 + G_{12} \cdot V_2 + \cdots + G_{1m} \cdot V_m = I_1 \\ G_{21} \cdot V_1 + G_{22} \cdot V_2 + \cdots + G_{2m} \cdot V_m = I_2 \\ \quad \vdots \qquad\quad \vdots \qquad\qquad\qquad \vdots \qquad\quad \vdots \\ G_{m1} \cdot V_1 + G_{m2} \cdot V_2 + \cdots + G_{mm} \cdot V_m = I_m \end{cases}$$

where $G_{11}, G_{12}, \ldots, G_{mm}$ and $I_1, I_2, \ldots, I_m$ are constants.

Solve the system of equations for the m node voltages $V_1, V_2, \ldots, V_m$ using Gaussian elimination or some other method.

Example:

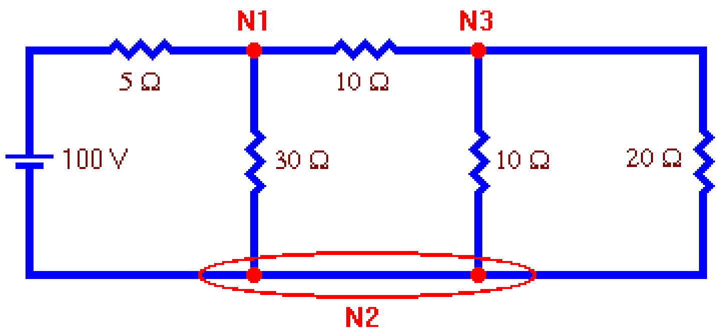
Use nodal analysis to find the voltage at each node of this circuit.



Solution:

Note that the "pair of nodes" at the bottom is actually 1 extended node.

Thus the number of nodes is 3.



We will number the nodes as shown above.

We will choose node 2 as the reference node and assign it a voltage of zero.

Write down Kirchoff's Current Law for each node. Call V_1 the voltage at node 1, V_3 the voltage at node 3, and remember that $V_2 = 0$. The result is the following system of equations:

$$\frac{V1}{30} + \frac{V1-100}{5} + \frac{V1-V3}{10} = 0$$

$$\frac{V3-V1}{10} + \frac{V3}{10} + \frac{V3}{20} = 0$$

node 1 and the second equation results from KCL applied at node 3. Collecting terms this becomes:

$$\left(\frac{1}{30}+\frac{1}{5}+\frac{1}{10}\right)V1-\left(\frac{1}{10}\right)V3=\frac{100}{5}$$

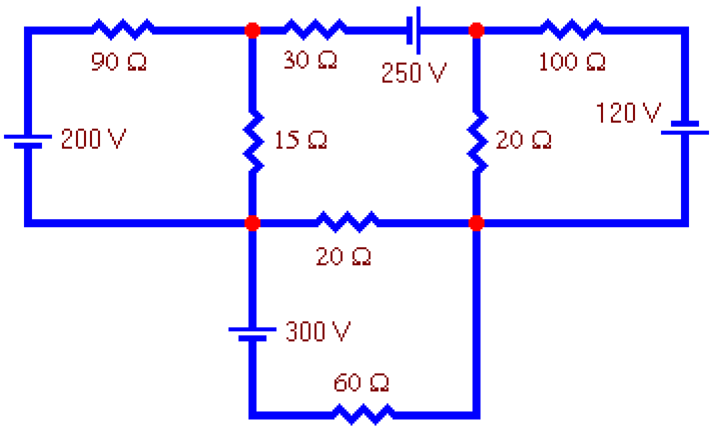
$$-\left(\frac{1}{10}\right)V1+\left(\frac{1}{10}+\frac{1}{10}+\frac{1}{20}\right)V3=0$$

This form for the system of equations could have been gotten immediately by using the inspection method.

Solving the system of equations using Gaussian elimination or some other method gives the following voltages: V_1=68.2 volts and V_3=27.3 volts

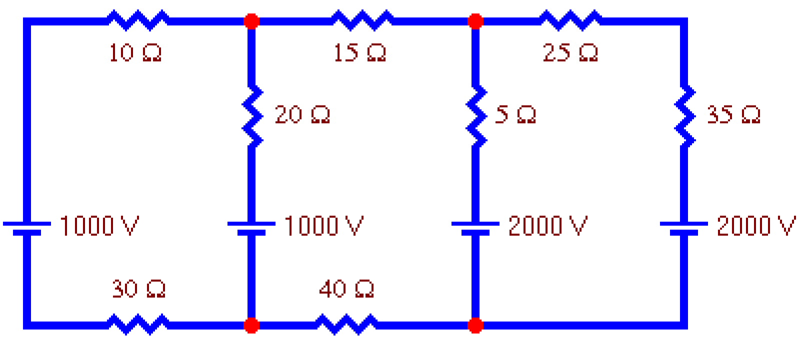
Exercise 2:

Use nodal analysis to find the voltage at each node of this circuit.



Exercise 3:

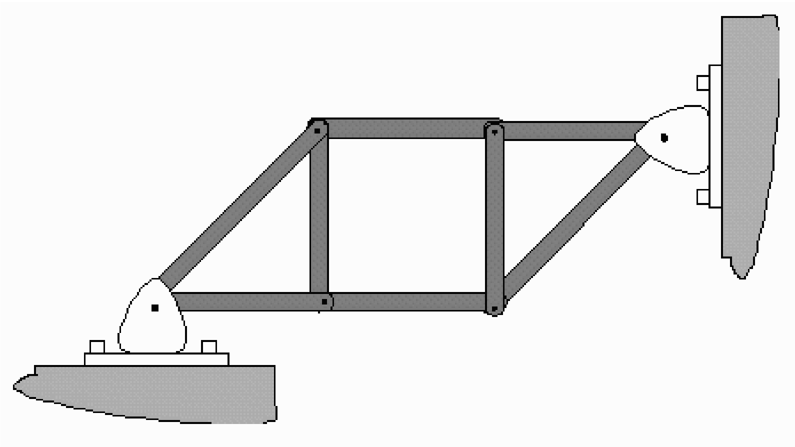
Use nodal analysis to find the voltage at each node of this circuit.



LINEAR ALGEBRA IN CIVIL ENGINEERING APPLICATION

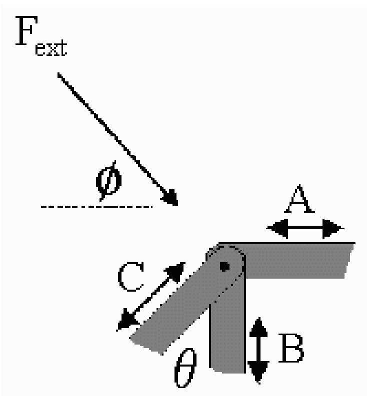
Truss Analysis:

Linear Algebra is used quite heavily in Structural Engineering. This is for a very simple reason. The analysis of a structure in equilibrium involves writing down many equations in many unknowns. Often these equations are linear, even when material deformation (i.e. bending) is considered. This is exactly the sort of situation for which linear algebra is the best technique. Consider, for example, the following two dimensional truss:



The beams are joined together by smooth pins, and the supports are fastened so that they cannot be moved. A simple analysis, using the method of joints, assumes that external forces will only act at the joints, and that the beams are perfectly rigid. This allows for only longitudinal forces in the beams. Given these assumptions, the truss is stable if and only if the vertical and horizontal components of the forces at each joint sum to zero.

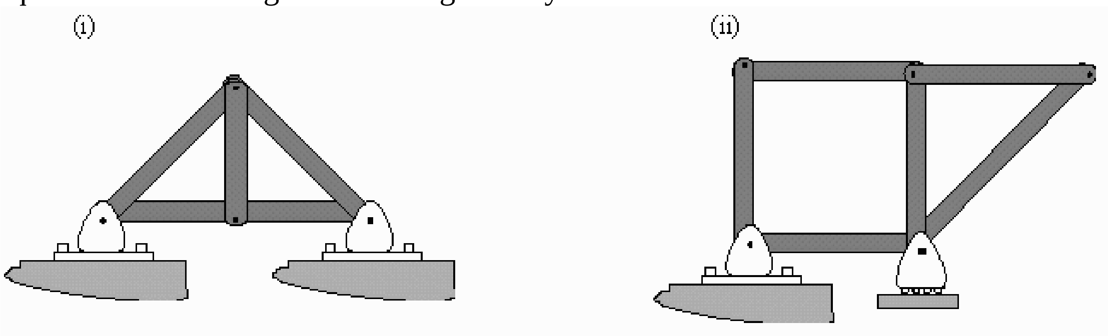
For example, in the joint shown to the left, the beams are under compression. Their internal forces, therefore, tend to act against the compression and out against the joint. Static equilibrium demands that $\sum_i F_{ix} = 0$ and $\sum_i F_{iy} = 0$. In particular, using the capital letters to denote the magnitude of force along a beam,



$$\text{Horizontal,} \quad -A + C(\sin\theta) + F_{ext}\cos\phi$$
$$\text{Vertical,} \quad B + C(\cos\theta) - F_{ext}\sin\phi$$

Because each joint will give such equations, the total number of equations that can be written, under our assumptions, is twice the number of joints. Normally, we are interested in finding the internal beam forces and the reaction forces at the support under some external load. *Thus the number of unknowns is the number of beam forces plus the number of reaction forces.*

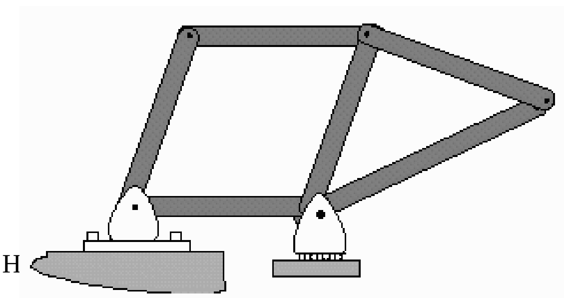
Here are some examples where we can say quite a bit about the truss only by counting equations and looking at the basic geometry:



The truss in case (i) is stable for certain. This is because the basic building blocks are triangular shapes. The triangle is a stable geometry in the sense that there exists only one unique triangle with the same dimensions (the Side-Side-Side Theorem). Thus, under our assumption of perfect rigidity, the truss cannot move even under some external load. There are 4 joints, and therefore 8 equations of static equilibrium. However, there are 5 beams and 4 reaction forces (normal forces fixing the supports), for a total of 9 unknowns. We there have *too few* equations to determine a unique solution. The truss is said to be indeterminate. To actually solve for the forces in the beams, the material properties of the beams have to be considered. This involves the stiffness of the beams as well as the possible existence of bending moments. Analysis of this truss is thus beyond our simple static equilibrium analysis.

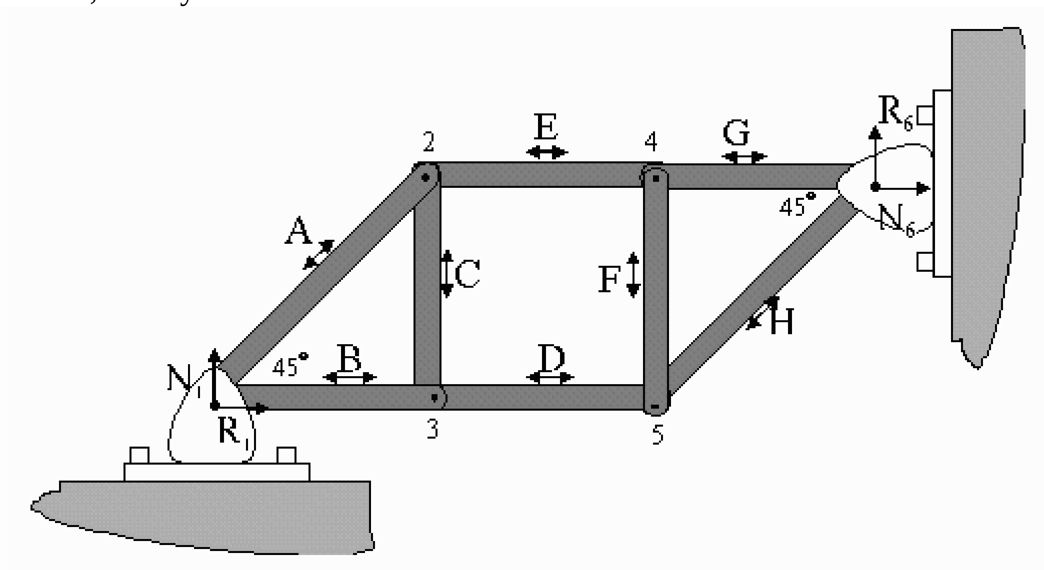
The truss in case (ii) is a different story. Since it is not constructed solely from triangles, there is no guarantee that it is geometrically stable. Moreover, there are 10 equations (from 5 joints), 6 beams and at most 3 reaction forces (the roller can only applies a normal force in the up direction). Thus even if the net external force is in the down direction, the system of equations consists of 10 equations with 9 unknowns. This system is over-determined and, in general, will not yield a solution. This is because there are not enough 'variables' to compensate and resist every conceivable configuration of forces.

For example, a little horizontal push on the leftmost beam will result in the collapse of the truss (see diagram).



Is the First Truss Stable?

Let’s get back to the first truss and analyze it. It is quickly apparent that it has the right number of equations to match the number of unknowns. Because it is not made up solely of triangles, the stability of the truss is not a clear matter. To analyze, we subject the truss to some hypothetical load, write down the static equilibrium set of equations, and try to solve.



Assuming compressive forces, as shown, and some kind of arbitrary loads at each joint, this is the set of equations that results:

1) $\rightarrow \quad R_1 - \frac{1}{\sqrt{2}} A - B = F_{1x}$

$\uparrow \quad N_1 - \frac{1}{\sqrt{2}} A = F_{1y}$

2) $\rightarrow \quad \frac{1}{\sqrt{2}} A - E = F_{2x}$

$\uparrow \quad \frac{1}{\sqrt{2}} A + C = F_{2y}$

3) $\rightarrow \quad B - D = F_{3x}$

$\uparrow \quad -C = F_{3y}$

4) $\rightarrow \quad E - G = F_{4x}$

$\uparrow \quad F = F_{4y}$

5) $\rightarrow \quad D - \frac{1}{\sqrt{2}} H = F_{5x}$

$\uparrow \quad -F - \frac{1}{\sqrt{2}} H = F_{5y}$

6) $\rightarrow \quad N_6 + G + \frac{1}{\sqrt{2}} H = F_{6x}$

$\uparrow \quad R_6 + \frac{1}{\sqrt{2}} H = F_{6y}$

The above system of equations can be written in matrix form as

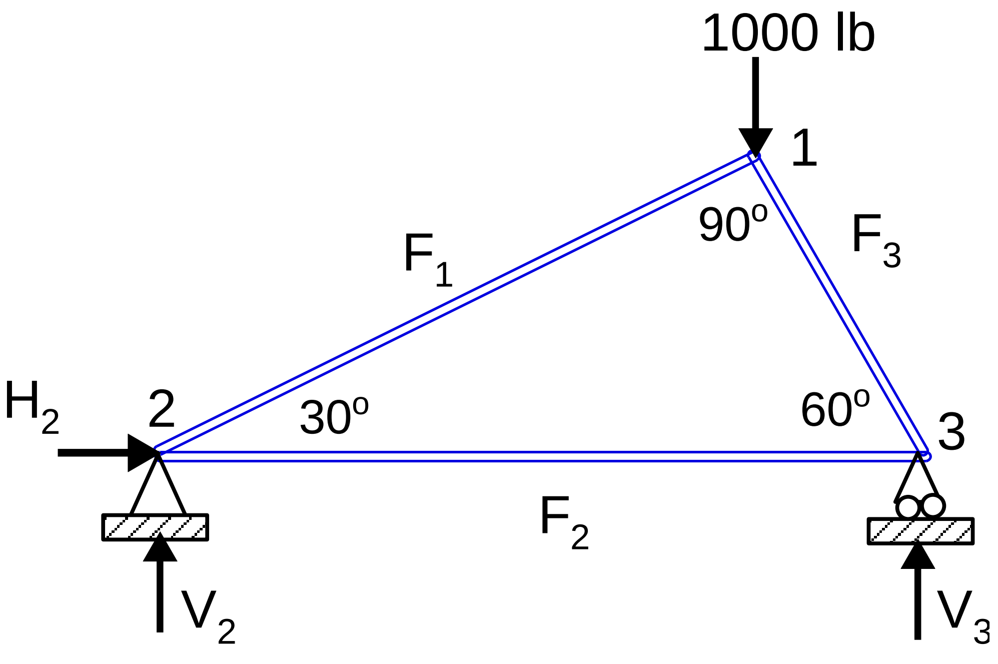
$$\begin{bmatrix}
1 & 0 & -1/\sqrt{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & -1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1/\sqrt{2} & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1/\sqrt{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1/\sqrt{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1/\sqrt{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1/\sqrt{2} & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/\sqrt{2} & 0 & 1
\end{bmatrix} \cdot \begin{bmatrix} R_1 \\ N_1 \\ A \\ B \\ C \\ D \\ E \\ F \\ G \\ H \\ N_6 \\ R_6 \end{bmatrix} = \begin{bmatrix} F_{1x} \\ F_{1y} \\ F_{2x} \\ F_{2y} \\ F_{3x} \\ F_{3y} \\ F_{4x} \\ F_{4y} \\ F_{5x} \\ F_{5y} \\ F_{6x} \\ F_{6y} \end{bmatrix}$$

This can be solved for any arbitrary external force configuration if and only if we can invert the matrix, *A*, on the left hand side. Notice that this matrix depends only on the geometry of the truss and not on the load.

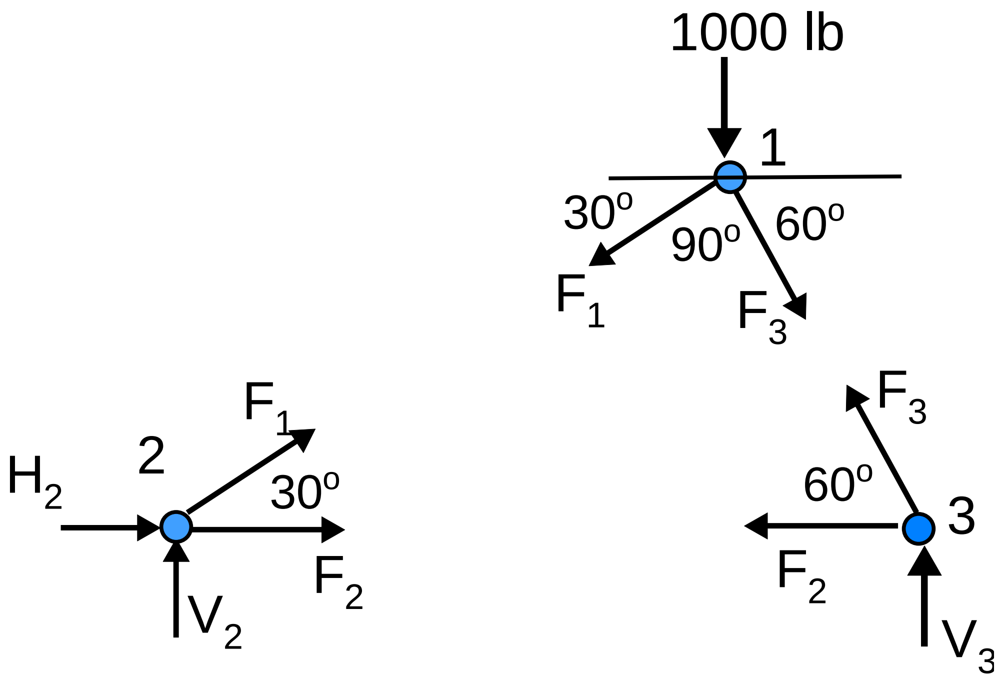
$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{2} & 0 & \sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

The fact that this matrix exists assures us that the truss is stable.

In Class Exercise:
Obtain the system of coupled linear algebraic equations describing the truss shown:



The truss free-body diagram is given below, try to obtain the system of linear equations describing the system



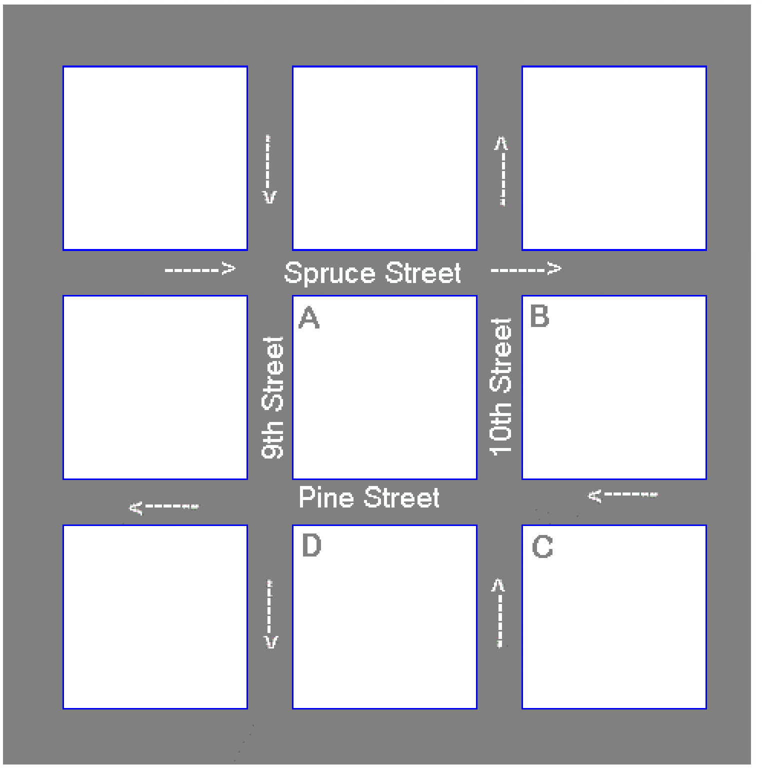
Answer:

$$F_1 = -500,\ \ F_2 = 433,\ \ F_3 = -866$$
$$H_2 = 0,\qquad V_2 = 250,\ \ V_3 = 750$$

APPLICATION TO TRAFFIC FLOW

Problem 1

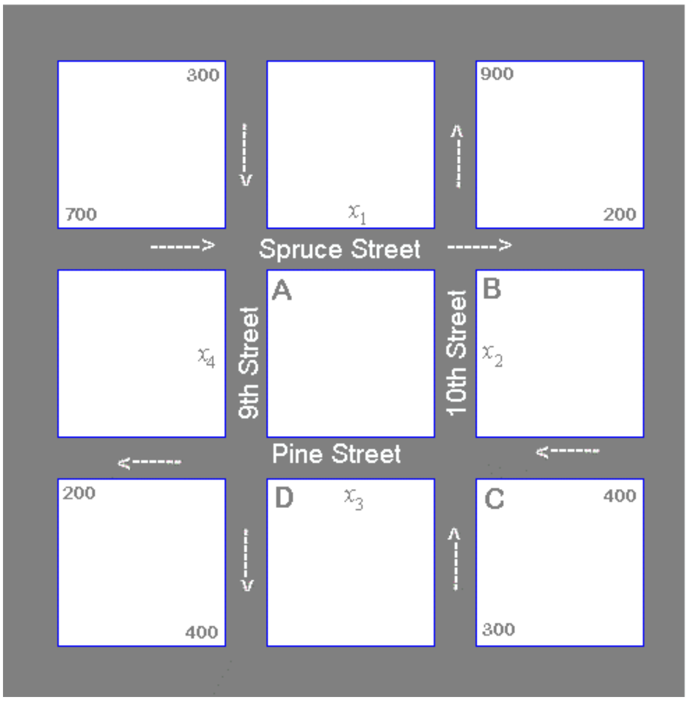
At rush hours, traffic congestion is encountered at the street intersections shown in the figure below. The city wishes to improve the traffic signals at these corners to improve the traffic flow. All streets are one-way and the directions are indicated by the arrows.



Data Collection. The traffic engineers gathered the following information

1. *Corner A*
   700 cars an hour come down Spruce Street to intersection A.
   300 cars an hour come down 9th Street to intersection A
2. *Corner B.*
   200 cars an hour leave intersection B on Spruce Street.
   900 cars an hour leave intersection B on 10th Street.
3. *Corner C.*
   400 cars an hour enter on Pine Street to intersection C.
   300 cars an hour come down 10th Street to intersection C.
4. *Corner D.*
   200 cars an hour leave intersection D on Pine Street.
   400 cars an hour leave intersection D on 9th Street to intersection A

Introduction of Labels.



Let x_1 denote the number of cars leaving corner A on Spruce Street towards corner B.
Let x_2 denote the number of cars arriving to corner B on 10th Street from corner C.
Let x_3 denote the number of cars leaving corner C on Pine Street towards corner D.
Let x_4 denote the number of cars arriving to corner D on 9th Street from corner A.

Assumptions. To solve this problem. we assume the following:

1. To speed the traffic flow every car that arrives to a given corner must also leave, hence at any corner, the number of cars arriving is equal to number of cars leaving.
2. All streets are one-way.
3. All variables, x_1, x_2 , x_3 and x_4 , are positive integers since they represent numbers of cars.

Equations. Using assumption 1 for each corner, we obtain the following equations:

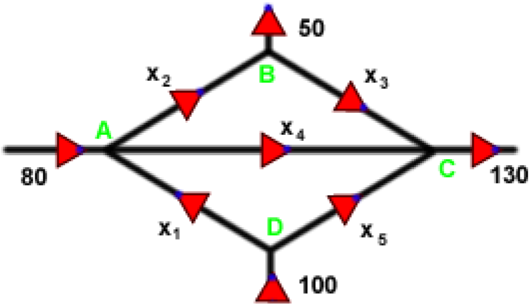
At corner A	*At corner B*	*At corner C*	*At corner D*
$x_1 + x_4 = 700 + 300$	$x_1 + x_2 = 900 + 200$	$x_2 + x_3 = 400 + 300$	$x_3 + x_4 = 400 + 200$

These four equations form a system of linear equation that can be solve using the Gauss-Jordan method (row reduction of the augmented matrix).

$$\begin{aligned} x_1 \qquad\quad &+x_4 = 1000 \\ x_1 + x_2 \qquad &\quad\;\; = 1100 \\ x_2 + x_3 &\quad\;\; = 700 \\ x_3 &+ x_4 = 600 \end{aligned}$$

Problem 2

Find all the x's in the following traffic network



Solution

set up a linear system describing how many enter and exit at each intersection (A, B, C, D)

- A: $80 = x_2 + x_4 - x_1$
- B: $50 = x_2 - x_3$
- C: $130 = x_3 + x_4 + x_5$
- D: $100 = x_1 + x_5$

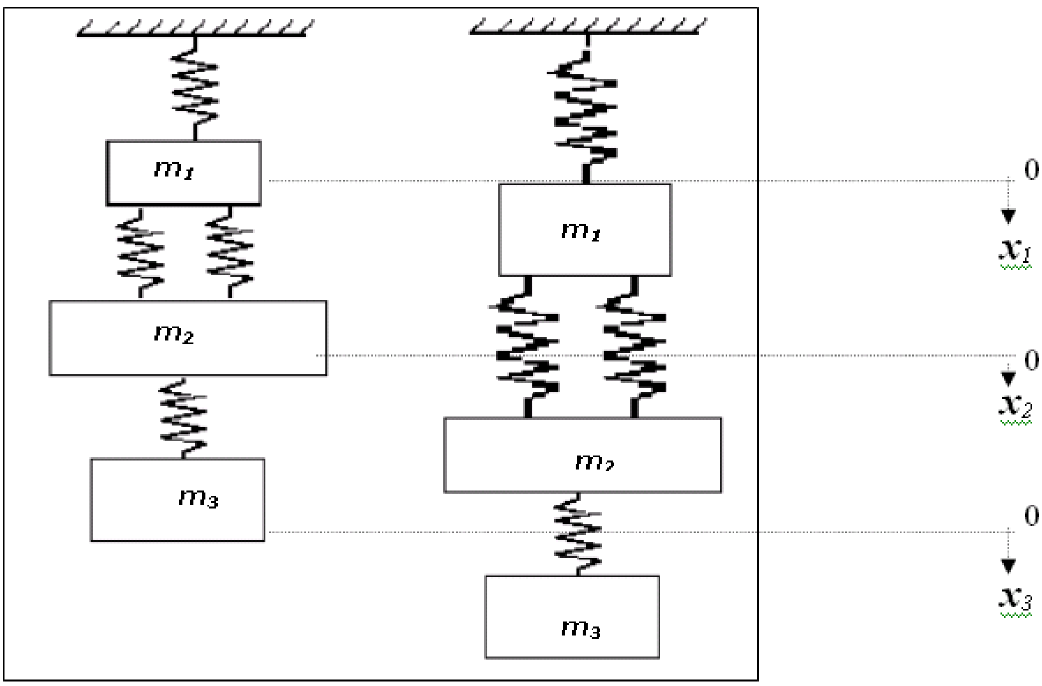
LINEAR ALGEBRA IN MECHANICAL ENGINEERING APPLICATIONS

Spring Mass Systems

Idealized spring-mass systems have numerous applications throughout engineering and linear algebra can successfully be applied in order to solve the problems associated with such systems.

Spring-mass systems play an important role in mechanical and other engineering systems. Such a system is shown in the figure. It composed of three masses, suspended vertically by a series of spring. The left portion of the diagram indicates the state of the system before release (i.e., the condition in which spring are neither stretched nor compressed).

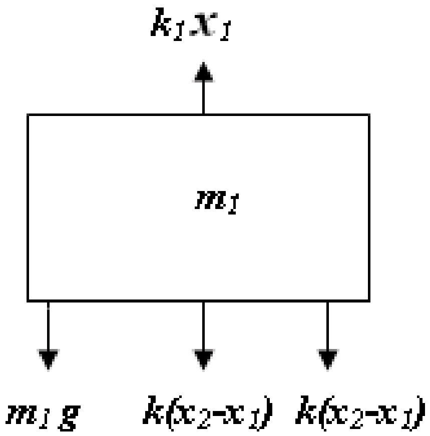
However, after the masses are released, they are pulled downward by the force of gravity. The resulting displacement of each spring is measured with respect to along local coordinates referenced to its initial position, as shown on right side of the diagram.



For each mass, Newton's second Law of motion (i.e., F=ma) can be applied in conjunction with force balances to develop the mathematical model of the system:

$$m\frac{d^2x}{dt^2} = F_D - F_U$$

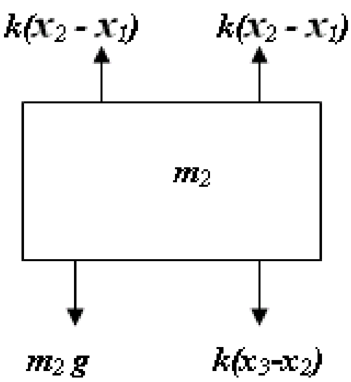
In order to simply the analysis, we'll assume that all spring are identical and follows Hook's Law. The free body diagram for first the first mass is shown in the figure:

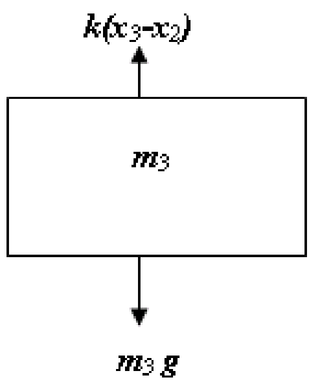


Therefore, net force acting on Mass m_1:

$$m_1\frac{d^2x}{dt^2} = m_1 g + 2\,k(x_2 - x_1) - kx_1$$

Thus, we have derived a second order ordinary differential equation to describe the displacement of the first mass with respect to time. However, it can be noticed that solution cannot be obtained because the model includes a second dependent variable x_2. Consequently, free body diagrams must be developed for the masses m_2 and m_3:





The net force acting on masses m2 and m_3 can be expressed by the following expression:

$$m_2 \frac{d^2x}{dt^2} = m_2 g + k(x_3 - x_2) - 2k(x_2 - x_1)$$

$$m_3 \frac{d^2x}{dt^2} = m_3 g - k(x_3 - x_2)$$

With appropriate initial conditions, these equations can be used to solve for the displacements of the masses as a function of time (i.e., their oscillations). For example, we have to find out the displacement when the system eventually comes to rest. To do this derivates, are set to zero:

$$\begin{aligned} 3kx_1 - 2kx_2 \qquad\quad &= m_1 g \\ -2kx_1 + 3kx_2 - kx_3 &= m_2 g \\ -kx_2 + kx_3 &= m_3 g \end{aligned}$$

Or, in Matrix form:

$$[K][X] = [W]$$

$$[X] = [K]^{-1}[W]$$

where [X] and [W] are the column vectors of the unknown X and the weight mg respectively. [K] is called stiffness matrix, is

$$[K] = \begin{pmatrix} 3k & -2k & \\ -2k & 3k & \\ & -k & -k \end{pmatrix}$$

Now, If $m_1 = 2$ kg, $m_2 = 3$ kg, $m_3 = 2.5$ kg, **k's** = 10 kg/sec^2

$$[X] = \begin{pmatrix} 30 & -20 & 0 \\ -20 & 30 & -10 \\ 0 & -10 & 10 \end{pmatrix} \begin{pmatrix} 19.6 \\ 29.4 \\ 24.5 \end{pmatrix}$$

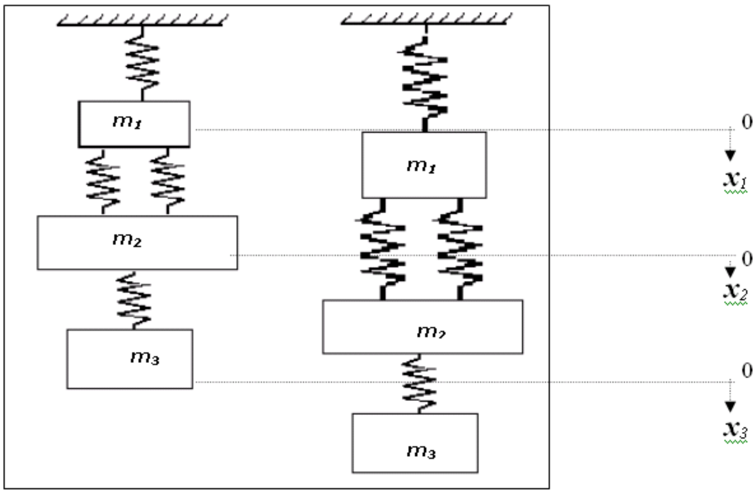
$$[X] = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7.350 \\ 10.045 \\ 12.495 \end{pmatrix}$$

$$[K]^{-1} = \begin{pmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.15 & 0.15 \\ 0.1 & 0.15 & 0.25 \end{pmatrix}$$

Each Element of this matrix k^{-1} tell us the displacement of mass *i* due to a unit force imposed on mass *j*.

Example:

Consider the following three masses and four spring's system

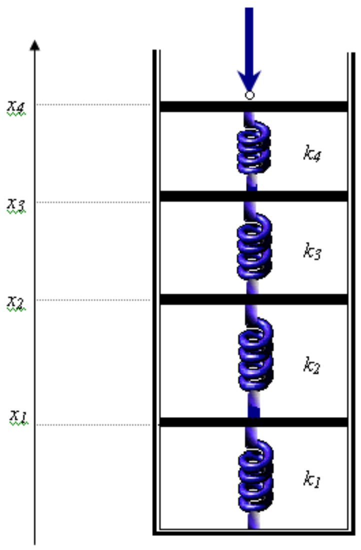


1- If m_1=10 kg, m_2=2 kg, m_3=2.0 kg and k's= 10 kg/sec^2. Compute the displacement x_1, x_2, x_3 and generate the inverse of K.

2- Add a third spring between masses m_1 and m_2 and perform the same computation.

Example:

Idealized spring-mass systems have numerous applications throughout engineering. In the figure shown below, an arrangement of four springs in series being depressed with a force of 200 lb is given. At equilibrium, force-balance equations can be developed defining the interrelationship between the springs:



$k_2(x_2\text{-}x_1) = k_1x_1$

$k_3(x_3\text{-}x_2)=k_2(x_2\text{-}x_1)$

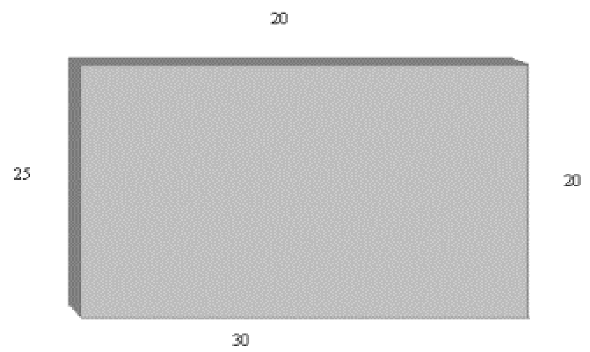
$k_4(x_4 - x_3)=k_3(x_3\text{-}x2)$

$F=k_4(x_4\text{-}x_3)$

Where the k's are spring constant. If k_1 through k_4 are 100, 50, 75 and 200 lb/in respectively, compute the x's.

Application to Temperature Distribution

Consider the cross section of a long rectangular metallic plate. As you can imagine, the boundaries of the plate are subject to three different temperatures, the following diagram represents this situation:



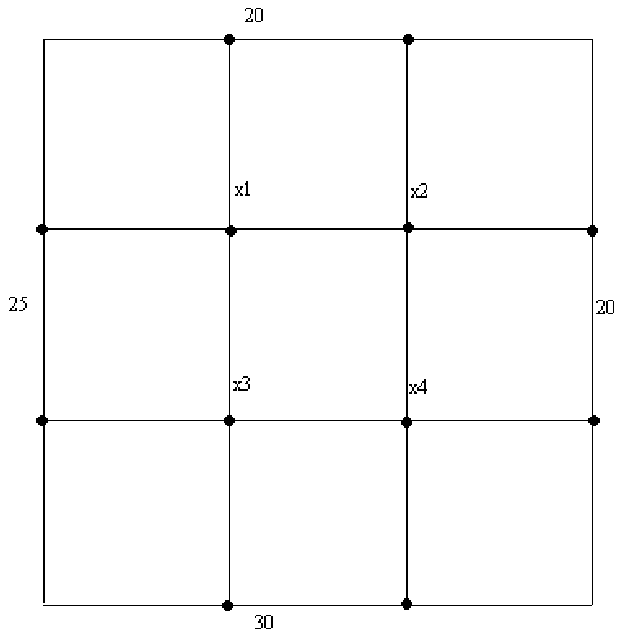
where the number represent the temperatures (in degree Celsius) of the boundaries. Engineers are interested in knowing the temperature distribution inside the plate in a specific period of time so they can determine the thermal stress to which the plate is subjected. Assuming the boundary temperatures are held constant during that specific period of time, the temperature inside the plate will reach certain equilibrium after some time has passed. Finding this equilibrium temperature distribution at different points on the plate is desirable, but extremely difficult. However, one can consider a few points on the plate and **approximate** the temperature of these points. This approximation is based on a very important physical property called the **Mean-Value Property:**

If a plate has reached a thermal equilibrium, and P is a point on the plate, C is a circle centered at P and fully contained in the plate, then the temperature at P is the average value of the temperature function over C.

To see how the property works, place a grid over the plate and consider the points where the lines of the grid meet. We will be interested in the temperatures at these points only in the plate. Design the grid in such a way that some of the points considered lie on the boundary of the plate. Studying the temperature at these grid points requires the following practical version of the Mean-Value Property:

If a plate has reached a thermal equilibrium and P is a grid point not on the boundary of the plate, then the temperature at P is the average of the temperatures of the four closest grid points to P.

Let us start with a grid with four interior points, and let x_1, x_2, x_3, x_4 be the temperatures at these four points. The situation is illustrated in the following diagram:



By the second version of the Mean-Value Property, we have the following system of linear equations:

$$
\begin{aligned}
x_1 &= \frac{20+25+x_2+x_3}{4}\\
x_2 &= \frac{20+20+x_1+x_4}{4}\\
x_3 &= \frac{25+30+x_1+x_4}{4}\\
x_4 &= \frac{20+30+x_2+x_3}{4}
\end{aligned}
$$

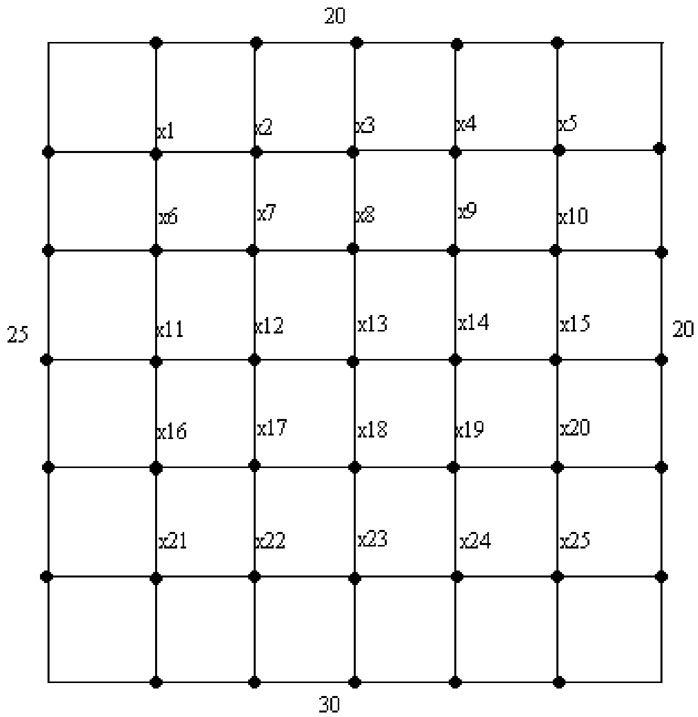
Rearrange the above equations to get a system of linear equations and find the solution (Temperature distribution inside the plate)

The matrix form of the system is $AX=b$, where

X is called the **vector of equilibrium temperatures.**

Suppose now that the boundary temperatures change from 25, 20 and 30 to 15, 10 and 20 respectively,
Find the new linear system matrix,
Find the new temperature distribution

The equilibrium temperature approximations found above could be more accurate if we consider a finer grid (more interior points). Consider the following grid obtained from the first one by reducing the spacing in grid 1 to half:



The new grid has 25 interior points. Repeat the same process as in the case of 4 interior points above and obtain the system matrix (try it yourself):

Solving for the Temperature distribution, use the first boundary conditions:

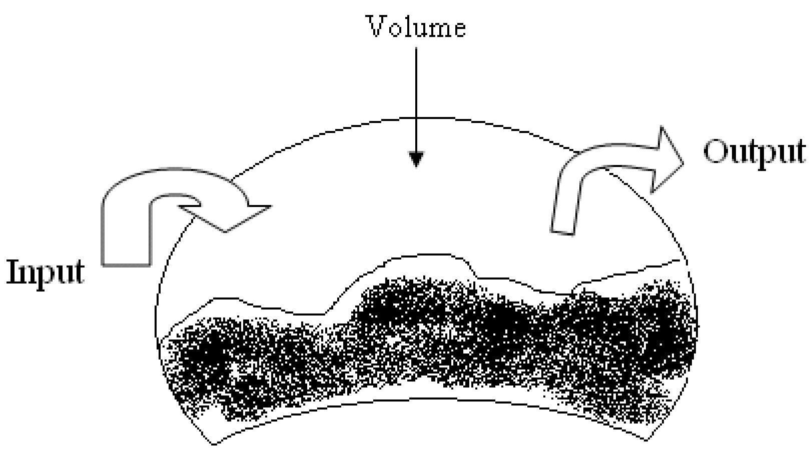
LINEAR ALGEBRA IN CHEMICAL ENGINEERING APPLICATIONS

One of the most important organizing principles in chemical engineering is the 'Conservation of Mass'. In quantitative terms, the principle is expressed as a mass balance that accounts for all sources and sinks of material that pass in and out of a unit volume. Over a finite number of time, this can be expressed as

Accumulation = Inputs – Outputs

The mass balance represents a bookkeeping exercise for a particular substance being molded. For the period of computation, if the inputs are greater than the outputs, the mass of a substance within the volume increases. If the outputs are greater than the inputs, the mass decreases. If inputs are equal to outputs, the accumulation is zero and mass remains constant. For this stable condition (i.e., steady-state, it can be represented as:

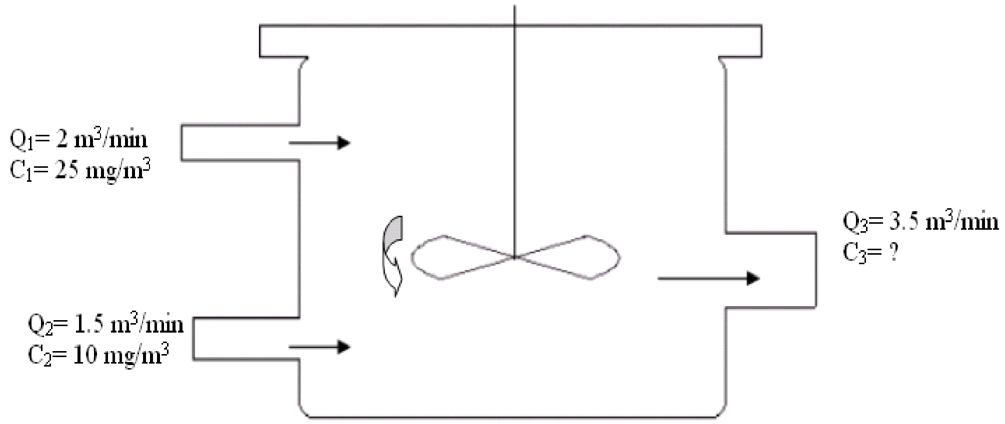
Input = output



Example:

Suppose we are performing a mass balance for a conservative substance (i.e., one that doesn't increase or decrease due to chemical transformation) in a reactor, we would have to quantify the rate mass flows into the reactor through the two inflow pipes and out of the reactor through outflow pipe. This can be done by thinking of Product flow rate, Q_1 (in m^3/min) and the concentration, c (in mg/m^3) for each pie. For example, for pipe 1, $Q_{1=}2$ m^3/min and C=25 mg/m^3 ; therefore, the rate at which mass flows into the reactor through pipe 1 is Q_1C = 50 mg/min.

Similarly, for pipe 2, mass flow rate can be calculated as Q_2C = 15 mg/min. Note that we have to find out the concentration of the reactor through pipe 3 and we have already sufficient information to calculate it.



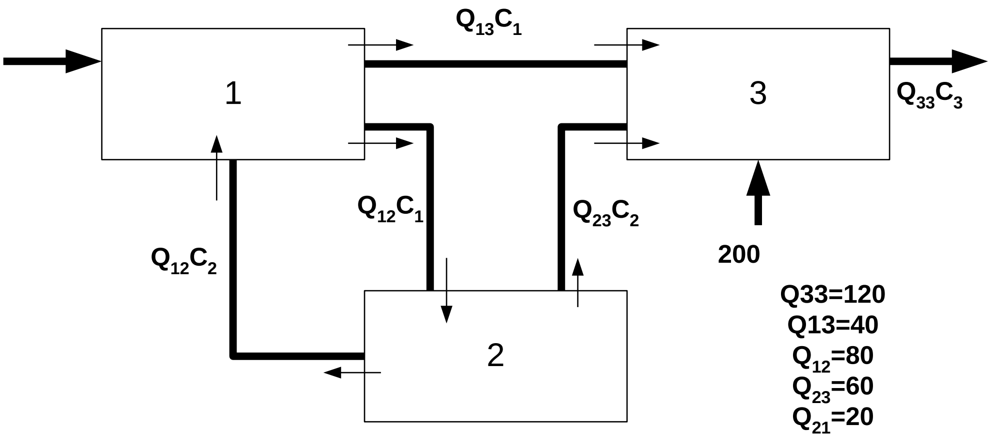
Because the reactor is at steady state, therefore:

$Q_1C_1 + Q_2C_2 = Q_3C_3$
$50 + 15 = 3.5\ C_3$
Therefore, C_3 =18.6 mg/m^3

Which is the concentration in the third pipe. Because, the reactor is well mixed (represented by the propeller), the concentration will be uniform (homogeneous) throughout the tank. Therefore, the concentration in pipe 3 should be identical to the concentration throughout the reactor.

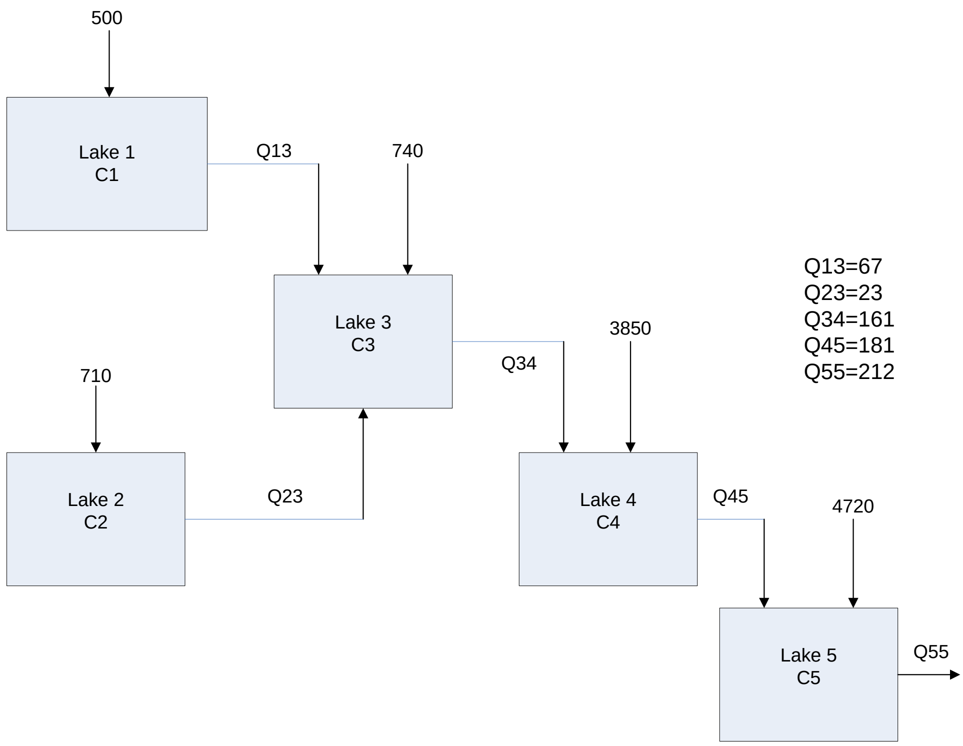
Exercise-1:

In the following figure, three reactors linked by pipes are shown. The rate of transfer of chemicals through each pipe is equal to a flow rate (Q, with units of cubic meter per second) multiplied by the concentration of the reactors from which the flow originates (i.e., with units of mg per cubic meter). If the system is at a steady state, the transfer into each reactor will balance the transfer output. Develop mass-balance equations and solve the three simultaneous linear algebraic equations for their concentrations.



Exercise-2:

Using the information shown in the figure below, determine the concentration of Chloride in each of the 5 Lakes.



APPLICATIONS TO CHEMISTRY

Application 1

It takes three different ingredients A, B, and C, to produce a certain chemical substance. A, B, and C have to be dissolved in water separately before they interact to form the chemical. Suppose that the solution containing A at 1.5 g/cm^3 combined with the solution containing B at 3.6 g/cm^3 combined with the solution containing C at 5.3 g/cm^3 makes 25.07 g of the chemical. If the proportion for A, B, C in these solutions are changed to 2.5, 4.3, and 2.4 g/cm^3, respectively (while the volumes remain the same), then 22.36 g of the chemical is produced. Finally, if the proportions are 2.7, 5.5, and 3.2 g/cm^3, respectively, then 28.14 g of the chemical is produced. What are the volumes (in cubic centimeters) of the solutions containing A, B, and C?

Solution Let x, y, z be the corresponding volumes (in cubic centimeters) of the solutions containing A, B, and C. Then 1.5x is the mass of A in the first case, 3.6y is the mass of B, and 5.3z is the mass of C. Added together, the three masses should give 25.07 g. So

$$1.5x + 3.6y + 5.3z = 25.07$$

The same reasoning applies to the other two cases. This gives the linear system

$$\begin{cases} 1.5x + 3.6y + 5.3z = 25.07 \\ 2.5x + 4.3y + 2.4z = 22.36 \\ 2.7x + 5.5y + 3.2z = 28.14 \end{cases}$$

The augmented matrix of this system is

$$\begin{bmatrix} 1.5 & 3.6 & 5.3 & 25.07 \\ 2.5 & 4.3 & 2.4 & 22.36 \\ 2.7 & 5.5 & 3.2 & 28.14 \end{bmatrix}$$

Reducing the above matrix would give the solution

$$x = 1.5, \quad y = 3.1, \quad z = 2.2$$

Application 2

Another typical application of linear systems to chemistry is **balancing a chemical equation**. The rationale behind this is the **Law of conservation of mass** which states the following:

"*mass is neither created nor destroyed in any chemical reaction. Therefore balancing of equations requires the same number of atoms on both sides of a chemical reaction. The mass of all the* ***reactants*** *(the substances going into a reaction) must equal the mass of the* ***products*** *(the substances produced by the reaction).*"

As an example consider the following chemical equation

$$C_2H_6 + O_2 \rightarrow CO_2 + H_2O.$$

Balancing this chemical reaction means finding values of x, y, z and t so that the number of atoms of each element is the same on both sides of the equation:

$$xC_2H_6 + yO_2 \rightarrow zCO_2 + tH_2O.$$

This gives the following linear system:

$$\begin{cases} 2x = z \\ 6x = 2t \\ 2y = 2z + t \end{cases}$$

The general solution of the above system is

$$\begin{cases} y = 7/2x \\ z = 2x \\ t = 3x \end{cases}$$

Since we are looking for whole values of the variables x, y z, and t, choose x=2 and get y=7, z= 4 and t=6. The balanced equation is then:

$$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O.$$

REVIEW QUESTIONS:

1. A total of \$50,000 is invested in three funds paying 6%, 8%, and 10% simple interest. The yearly interest is \$3,700. Twice as much money is invested at 6% as invested at 10%. How much was invested in each of the funds.

Answer: \$30,000 is invested at 6%, \$5,000 is invested at 8%, and \$15,000 is invested at 10%

2. Your company has three acid solutions on hand: 30%, 40%, and 80% acid. It can mix all three to come up with a 100-gallons of a 39% acid solution. If it interchanges the amount of 30% solution with the amount of the 80% solution in the first mix, it can create a 100-gallon solution that is 59% acid. How much of the 30%, 40%, and 80% solutions did the company mix to create a 100-gallons of a 39% acid solution?

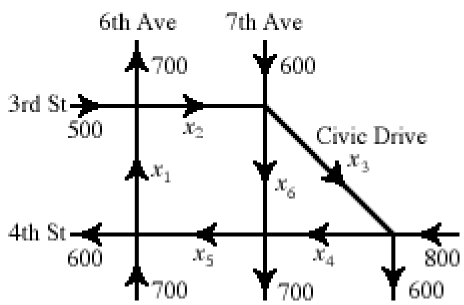
Answer: 50 gallons of the 30% solution, 40 gallons of the 40% solution, and 10 gallons of the 80% solution.

3. Five hundred tickets were sold for a certain music concert. The tickets for the adults sold for \$7.50, the tickets for the children sold for \$4.00, and tickets for senior citizen sold for \$3.50. The revenue for the Monday performance was \$3,025. Twice as many adult tickets were sold as children tickets. How many of each ticket was sold?

Answer: 300 adult tickets, 150 children tickets, and 50 senior citizen tickets.

4. A bakery displays the number of ounces of yogurt, wheat, and butter used in the production of one patch of its products. It uses 0.625 kg of yogurt, 0.625 kg of wheat and 0.625 kg of butter in a patch of rolls; 0.9375 kg of wheat and 0.9375 kg of butter in a patch of cookies; and 1.25 kg of yogurt and 1.25 kg of butter in a patch of bread. The bakery is supplied with 400 kg of yogurt, 350 kg of wheat, and 500 kg of butter, which must be used up completely.
   a. Put the above information in a table format.
   b. What is the maximum number of patches of all products that can be made to completely use up all the supplies?

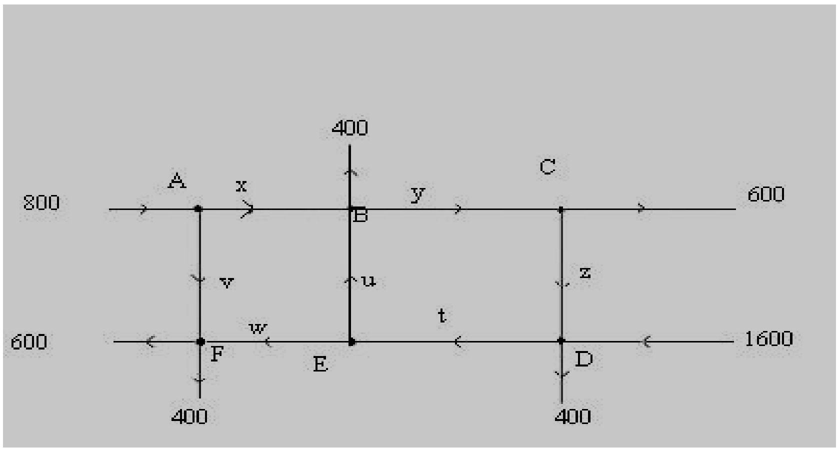
5. The flow of traffic in a downtown area is shown by the following diagram:



The numbers and letters represent the number of cars per hour using each section of road, while the arros indicate the direction that traffic flows on each section of road. A road section can handle up to 1000 cars per hour without congestion.

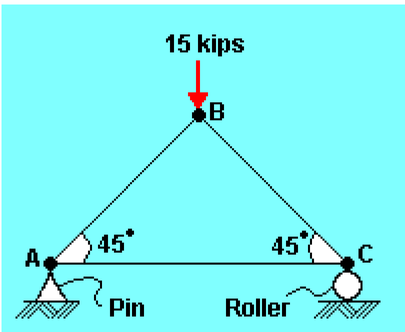
a. Set up a system of linear equations describing the traffic flow. (Hint: the number of cars going in to each intersection must equal the number of cars going out.)
b. Solve the system of equations, and give two solutions in which no road section is congested.
c. Suppose that the section of 7th Ave between 3rd and 4th streets is going to be closed for road work (so that $x_6 = 0$). Find flow pattern with this restriction which avoids congestion.

6. The diagram below represents the traffic flow through a certain block of streets. (The numbers are the average flows into and out of the network at peak traffic hours)



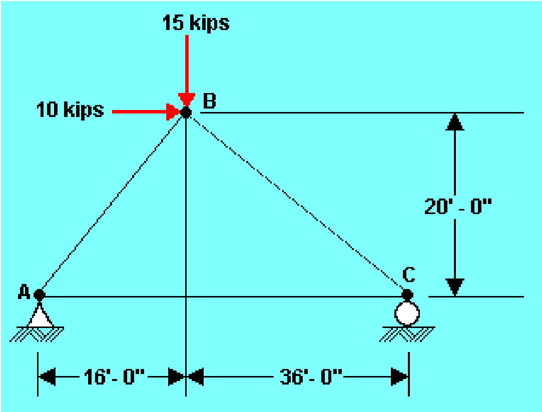
The flow into an intersection is equal to the flow out. Setup the system of equations and solve for traffic flow in all directions.

7. Calculate the forces in all members of the truss shown in the following diagram using the method of joints. [AB =10,600 lb (C), CB = 10,600 lb (C)]



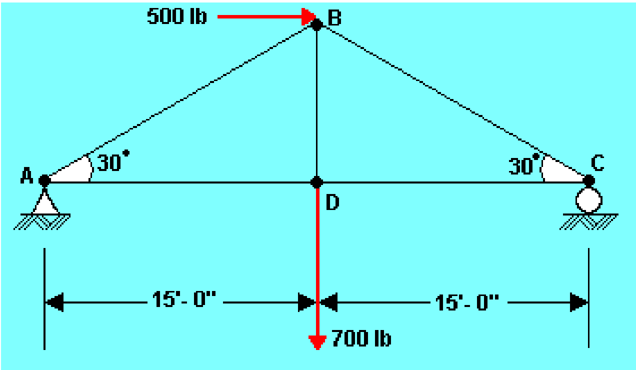
Problem #1

8. Calculate the forces in all member of the truss shown in the following diagram using the method of joints. [Ay = 6,540 lb., Ax = -10,000 lb, Cy = 8460 lb., AB = 8,375 lb, AC = 15,230 lb, BC=17,420 lb)



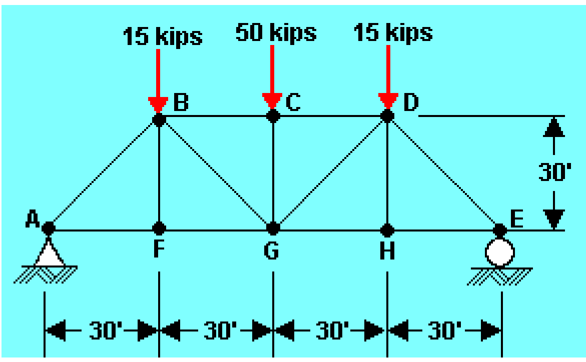
Problem #2

9. Calculate the forces in all members of the truss shown in the following diagram using the method of joints. (Ay = 225 lb, Ax = -500 lb., Cy = 475 lb., AB = 450 lb., AD = 890 lb., BC = 950 lb.)



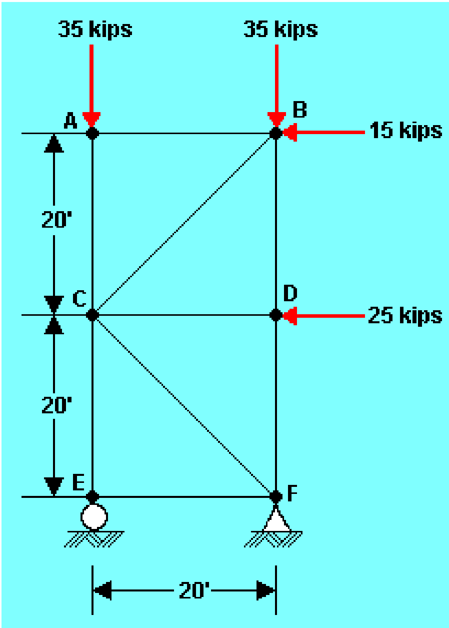
Problem #3

10. Calculate the forces in all members of the truss shown in the following diagram using the method of joints. [Ay=Ey=40,000 lb., AB =56,580 lb. (C), AF = FG = 40,000 lb. (T), BF=0, BC = CD =65,000 lb (T), BG = 35,360 (T), CG = 50,000 lb, right side same by symmetry]



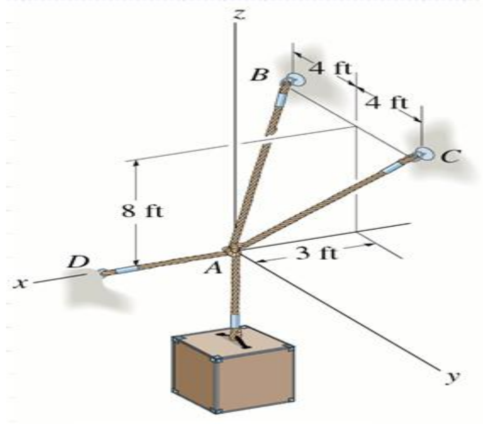
Problem #4

11. Calculate the forces in all members of the trusses shown in the following diagram using the method of joints. [Ey =90,000 lb., Fy=-20,000 lb., Fx=40,000 lb., AB = 0, AC = 35,000 lb. (C), BC = 21,220 lb. (C), BD=20,000 lb. (C), CD = 25,000 lb. (C), DF = 20,000 lb. (C), CE = 90,000 lb. (C), EF = 0, CF = 56,580 lb.(T)]

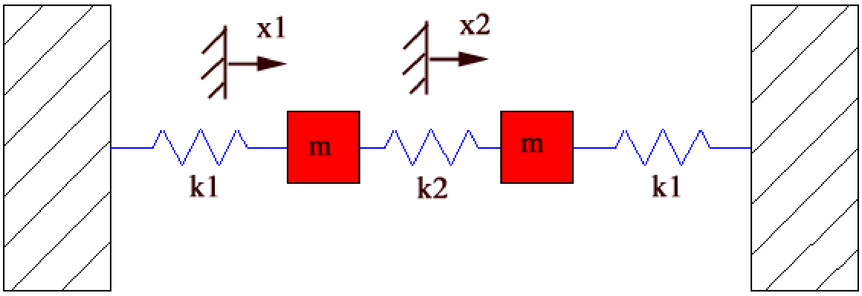


Problem #5

12. Determine the force in each cable used to support the 40 lb crate.

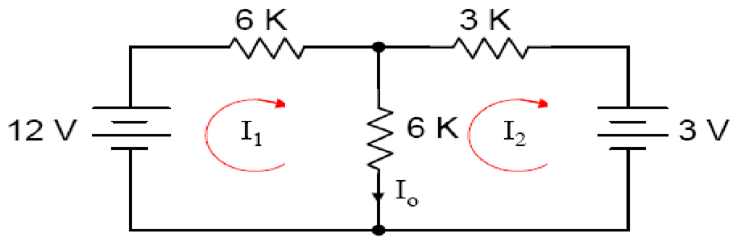


13. Consider the system shown below with 2 masses and 3 springs. The masses are constrained to move only in the horizontal direction (they can't move up an down):

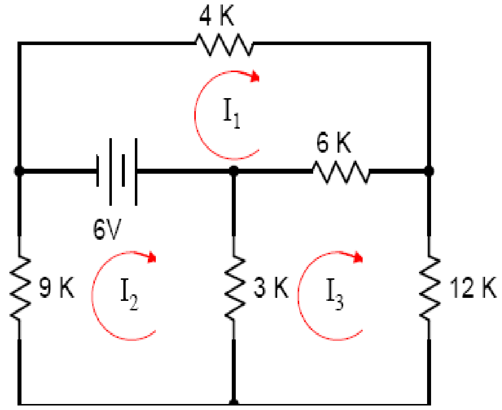


Use the solution we did in class to obtain the system response for K1=K2=2, and m=3. Use Matlab to plot the system response as done in the note

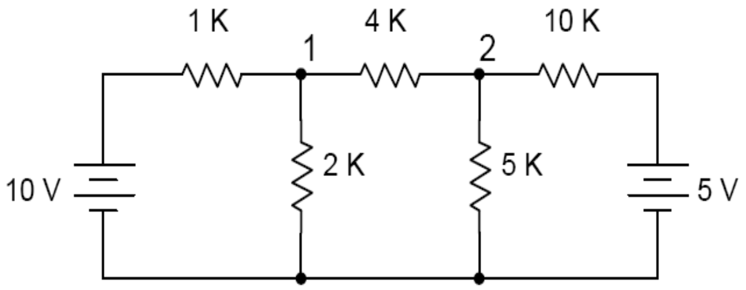
14. Use mesh analysis to calculate the current in each branch of the circuit given below.



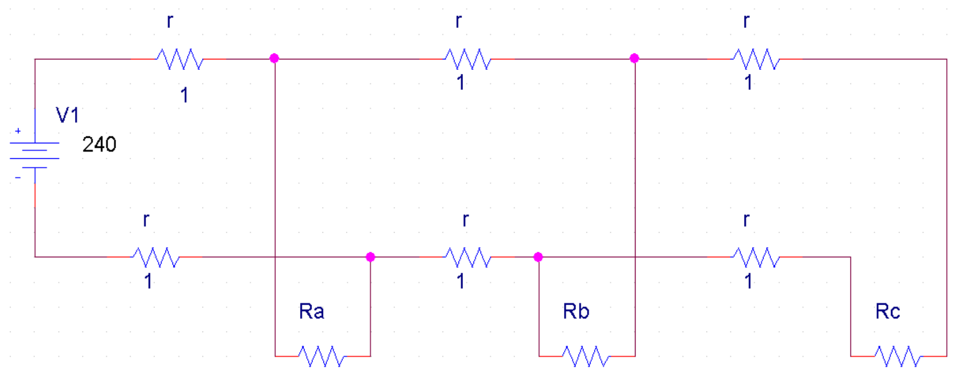
15. Use mesh analysis to calculate the current in each branch of the circuit given below



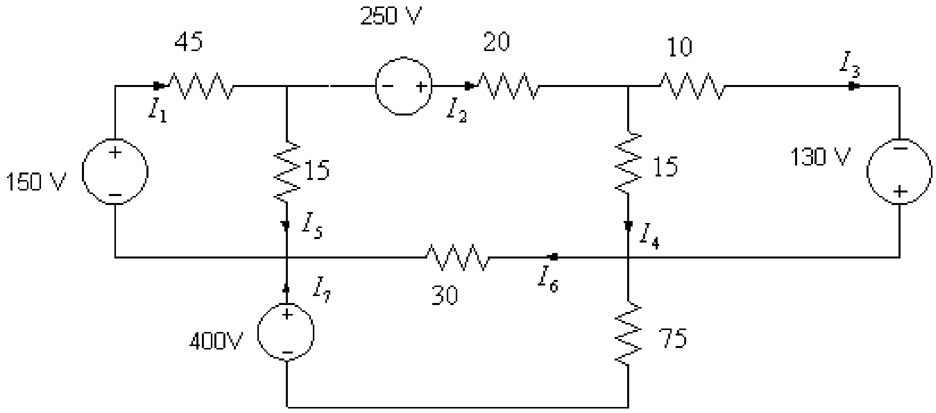
16. Use nodal analysis to calculate the current in each branch of the circuit given below



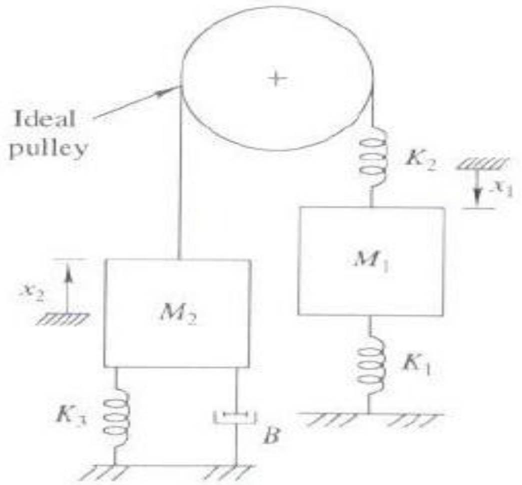
17. For the circuit given determine the current in Ra, Rb, and Rc, if all of them have a resistance of 500 kohm



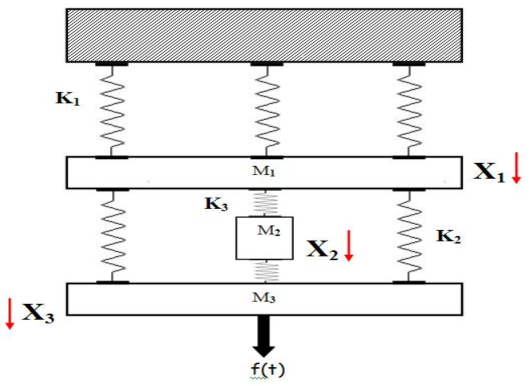
18. For the circuit shown use KVL to solve for the branch currents indicated, find branch currents (I_1-I_7)



19. For the system shown
    a. Draw Free-Body Diagram
    b. Write the modeling equations
    c. Compute the displacements x_1 and x_2 when the system eventually comes to rest

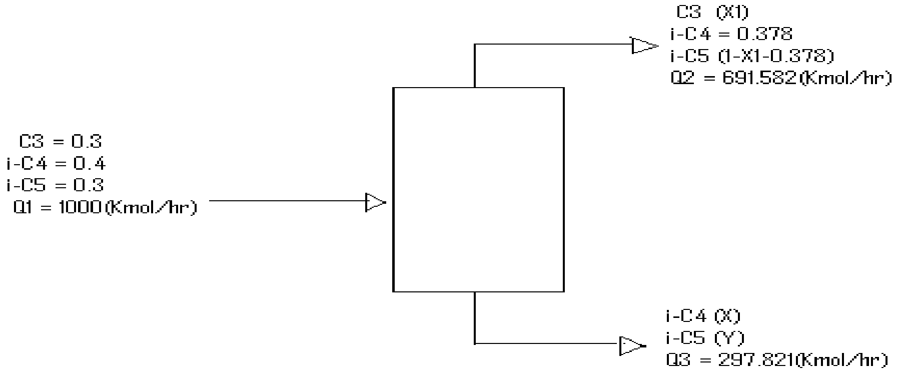


20. For the system shown
    a. Draw Free-Body Diagram
    b. Write the modeling equations
    c. Compute the displacements $x_{1,}$ x_2 and x_3 when the system eventually comes to rest

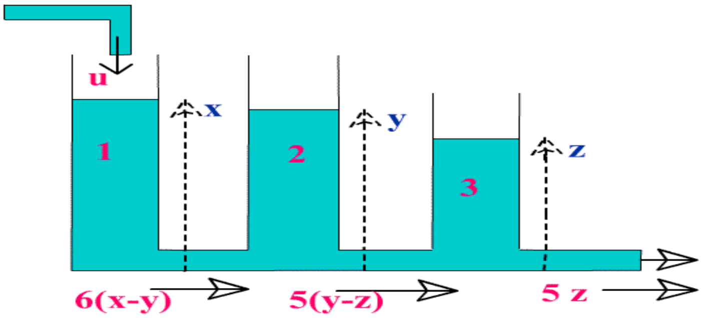


21. A hydrocarbon feed consisting of a mixture of propane, Isobutane, and Isopentane is fractionated at a rate of 1000(Kmol/hr) into a distillate that contains all the propane and 78% of the Isopentane in the feed. The mol

fraction of Isobutane in the distillate is 0.378. Given that the hydrocarbon feed from the distillate at a rate of 691.582(Kmol/hr) and feed from the bottom stream 297.821(Kmol/hr), determine the concentrations of the mixture component in the leaving streams.



22. In an industrial process of water flows through three tanks in succession as illustrated in the figure. The tanks have unit cross-section and have heads of water x, y and z respectively. The rate of inflow into the first tank is **u**, the flow rate in the tube connecting tanks 1 and 2 is **6(x-y)**, the flow rate in the tube connecting tanks 2 and 3 is 5(y-z), and the rate of outflow from tank 3 is **5z**. Assuming the system at stead state solve for the level of water (x, y, z) in each tank.



23. The action of sunlight on green plants powers a process called *photosynthesis*, in which plants capture energy from light and store energy in chemical form. The process is manifested as follow

$$CO_2 + H_2O \Rightarrow O_2 + C_6H_{12}O_6$$

Find the smallest positive integer to balance the equation.